Sanctions without Markups: Sourcing as the Margin of Bureaucratic Inefficiency⋆

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Abstract

When mandates secure compliance, what efficiency does the state forgo? Standard models of accountability (Prendergast, 2007) predict suppliers extract delivery-risk premia; passive-waste accounts (Bandiera et al., 2009) predict the buyer's sourcing pattern shifts. The two are observationally equivalent in cross-buyer data, which conflate within-firm pricing with equilibrium supplier composition. Using bid-level data on 479,330 pharmaceutical purchases by the São Paulo State Department of Health (2009–2019), we exploit a parallel administrative-request channel that shares all planning constraints of court-mandated procurement but carries no penalty for officials. Manski-Lee bounds place the selection-corrected "under-the-gun" gap at $[15.9\%, 21.1\%]$. *There is no within-firm markup* in deep markets ($\hat{\beta} = 0.035$, SE $= 0.041$ within firm-buyer-item triples), with a sanction

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premium reappearing on thin-supplier subsamples consistent with delivery-risk pricing. In deep markets, the price margin operates through demand fragmentation (admin orders are roughly 3.3× larger) and a sourcing shift (the modal winner differs across regimes in 70.2% of item-buyer pairs); a placebo on items never subject to litigation returns economically and statistically zero. Bounded welfare cost: $27.8 M per year on $300 M of annual litigated spending in São Paulo. The policy lever is demand aggregation, not contract design: delivery is guaranteed; sourcing efficiency is the bill.
Keywords: public procurement, health litigation, enforcement costs, judicial mandates, accountability design, sourcing

JEL: D44, D73, H51, H57, I18, K41

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1. Introduction

When mandates secure compliance, what efficiency does the state forgo? Standard models of bureaucratic accountability (Prendergast, 2007) predict suppliers extract delivery-risk premia; passive-waste accounts (Bandiera et al., 2009) predict the buyer's sourcing pattern shifts. The two predictions are observationally equivalent in cross-buyer data, where buyer-supplier match heterogeneity confounds within-firm and between-firm margins. We provide a within firm-buyer-item test in a setting where accountability pressure bites hardest: court-mandated pharmaceutical procurement in São Paulo under fines, personal liability, and fund seizure for delivery failure. The within-firm prediction fails; the sourcing prediction holds. Annual litigated pharmaceutical spending in São Paulo runs \$300 M; at stake is the procurement efficiency the state forgoes within that pool.

We use the universe of pharmaceutical procurement by the São Paulo State Department of Health (SES/SP) from 2009 through 2019: 479,330 purchase-offer-item observations on the state's electronic procurement platform (BEC), mandatory for common goods since 2007. The identification asset is institutional. Since 2009 the SES/SP scientific committee has run an administrative-request channel parallel to the courts, with the same compressed timelines (1–10 days), the same small quantities, the same diverted budget, and the same auction procedures on BEC. The two channels differ only in penalty exposure: failure to deliver under a court order triggers personal fines, liability, and fund seizure for the procurement officer; failure to deliver under the administrative channel triggers nothing. To our knowledge, no other Brazilian state operates a comparable dual channel for urgent pharmaceutical demand.

Three findings emerge. There is no within-firm markup in deep markets: within firm-buyer-item triples observed under both regimes (1,206 triples, 4,573 observations), the Admin coefficient is $\hat{\beta} = 0.035$ (SE 0.041), statistically indistinguishable from zero. The null is robust on above-median-quantity items, SUS-formulary items, and the later period; it breaks on thin-supplier subsamples (below-median quantity, earlier period), where the incumbent does extract a sanction premium consistent with Prendergast (2007). The deep-market within-firm null is the central finding; the thin-market deviation locates where the delivery-risk-premium prediction does bite. The bounded under-the-gun gap is modest: under Manski-Lee monotone restrictions on committee selection, the gap lies in $[15.9\%, 21.1\%]$, at the lower end of within-buyer procurement-cost dispersion documented in comparable settings (e.g., Best et al., 2023; Bosio et al., 2022). And the gap operates through demand fragmentation and a sourcing shift: administrative orders are roughly 3.3× larger than litigated orders; the bulk-discount channel mechanically delivers −32.8% in admin-minus-litigated log price. Among item-buyer pairs observed under both regimes, the modal winning firm differs across regimes in 70.2% of pairs; mean Jaccard similarity of the winner sets is 0.268. Sanctions reorganize the supplier set far more than they reorganize the incumbent's price (Figure 1, Table 5).

These findings depend on bounded admin-channel selection. The scientific committee admits on cost-effectiveness criteria, and a pre-period probit confirms that admin over-represents items that would have been cheaper under any regime: the coefficient on pre-period log reference price is −0.051 ($p < 0.001$), and on SUS-formulary status −0.265 ($p < 0.001$). The Lee bound corrects for this wedge.[1] Wild cluster bootstrap on the preferred-FE Admin coefficient gives $p_{\text{boot}} = 0.0080$ (999 Rademacher replicates; 95% log-point CI $[-0.469, -0.039]$). A Borusyak-Jaravel-Spiess event study with a Rambachan-Roth (Rambachan and Roth, 2023) honest-sensitivity overlay is

[1] A parametric Heckman correction is non-identified in this design (Online Appendix A.5).

supportive in the Appendix; the cross-sectional fixed-effects design is primary. A placebo on items never subject to litigation returns economically and statistically zero coefficients (−0.020, SE 0.032 for negotiated prices): the urgency premium exists only inside the litigation pool.

The central contribution is conceptual: identifying which margin of public procurement efficiency cedes under accountability that punishes non-delivery one-sidedly. The sourcing-vs-pricing distinction is the empirical lever—the cross-buyer literature cannot separate within-firm pricing from equilibrium supplier selection; the within firm-buyer-item triple does, and isolates sourcing under fragmentation as the empirical signature of accountability-induced inefficiency. Two supporting contributions follow. Whereas Coviello et al. (2018) document timeline *extension* as a procurement cost driver, we identify timeline *compression* under sanctions as an alternative channel, with the cost concentrated where demand aggregation matters most. The finding directly engages recent JPubE evidence on group purchasing in health (Lin and Wang, 2025), which identifies the same demand-aggregation lever we find compromised by court mandates. The paper engages procurement design and frictions (Bandiera et al., 2009; Best et al., 2023; Bosio et al., 2022; Decarolis et al., 2020; Carril et al., 2026), accountability-induced bureaucratic distortion (Prendergast, 2007; Bandiera et al., 2021), and the judicialization of health in Latin America (Wang, 2015; Ferraz, 2009; Biehl et al., 2009). Methodologically, the design honors post-treatment-bias (Acharya et al., 2016): quantity- and firm-fixed-effect controls are read as channelisolating rather than direct-effect estimators. Applied to the \$300 M of annual litigated spending in São Paulo with an admissibility calibration of 50.0%, the bounded UTG implies a per-unit-price welfare cost of \$27.8 M per year (Lee range $[\$23.9\ M, \$31.7\ M]$). The policy lever is demand aggregation, not contract design.

2. Institutional Background

The setting has three layers. A constitutional right to health generates mass litigation. A mandatory electronic procurement platform (BEC) records every transaction at bid level. And a parallel administrative-request channel, running alongside the courts since 2009, shares the litigated regime's planning constraints but spares the procurement officer from personal sanction. To our knowledge, no other Brazilian state and no Latin American jurisdiction we have surveyed operates a comparable dual channel for urgent pharmaceutical demand.

2.1. Right to Health and Mass Litigation

Brazil's 1988 Constitution declares health "a right of all and a duty of the state" (Article 196). Brazilian courts interpret Article 196 strictly: any individual may obtain virtually any medication through litigation, from common analgesics to rare-disease treatments costing over US\$400,000 per year.[2] Access is inexpensive—a prescription and a public defender suffice—

[2]Key STF rulings: Recurso Extraordinário 855.178/SE (Tema 793, 2015), establishing joint-and-several liability of federal entities for health-service provision; and Recurso Ex-

and approximately 85% of first-instance claims are granted (CNJ/INSPER, 2019). Litigated health spending in São Paulo amounts to approximately US$300 million annually, about 5% of the state public health budget.

2.2. Sanctions and the Administrative-Request Channel

The São Paulo State Department of Health (SES/SP) delivers health services through 99 decentralized public buyer units (PBUs). All purchases pass through BEC, mandatory for common goods since 2007, with two competitive procedures: *convite* (sealed-bid, smaller lots) and *pregão eletrônico* (multi-round descending auction, dominant for urgent purchases).[3] Ordinary purchases use a planning phase of 30–180 days, allowing demand aggregation across PBUs. 99.94% of court decisions in our sample arrive as preliminary injunctions with delivery deadlines of 1–10 days, compressing the internal phase to roughly one-third of the ordinary timeline.

The penalties for non-compliance are concrete and personal. The binding sanction is a daily fine (*astreinte*) calibrated to the purchase value, frequently reaching the order value within weeks of accumulation; additional sanctions include administrative and criminal liability for the responsible official and sequestration of state funds. The court order is referenced in the tender notice, so suppliers see the sanction context as they bid.

traordinário 566.471/RN (Tema 6, 2020) and Recurso Extraordinário 657.718/MG (Tema 500, 2019), defining conditions under which courts may order provision of high-cost and non-ANVISA medications respectively.

[3] Among the both-types cells that identify our tightest specifications, 95.5% of observations are processed through pregão eletrônico.

The institutional asset that identifies our comparison is the *administrative-request channel*, instituted by the SES/SP in 2009. Patients (or social workers acting on their behalf) request a medication directly through SES/SP intake before any litigation; a scientific committee evaluates the request under evidence-based-medicine and cost-effectiveness criteria. Admitted requests proceed through urgent procurement on BEC under the same compressed-timeline regime as litigated purchases. The relevant asymmetry is in the penalty regime: **failure to complete an administrative purchase carries no court order, no daily fine, no fund seizure, and no personal liability.** The supplier pool is identical: 92% of urgent winners also serve ordinary procurement of the same item-class. Between 2009 and 2019, the administrative channel generated 9,700 purchases, about 6.5% of the litigated volume; the smaller volume reflects committee gating, not procedural difference. We bound the selection-into-admin wedge formally in Section 4.

Table 1 summarizes the three regimes.

Table 1: Purchase types: institutional characteristics.

	Ordinary	Administrative	Litigated
Trigger	Standard demand	Patient request	Court order
Gatekeeper	PBU plan	SES/SP committee	Lower-court judge
Source of funds	Dedicated budget	Diverted budget	Diverted budget
Quantity	Large	Small	Small
Delivery deadline	30–180 d	1–10 d	1–10 d
Penalty for failure	None	None	**Daily fine, liability, fund seizure**
Visible to bidders	—	—	Court order in tender notice
Active since	Pre-2007	2009	1990s

Notes: Administrative and litigated purchases share all execution constraints (compressed timeline, small quantity, diverted budget, BEC platform, identical auction procedures) and differ only in the penalty regime visible to the procurement officer and to bidders. The within-urgent contrast we exploit in Section 4 compares these two columns.

3. Data and Sample

3.1. Sources

Three administrative datasets, January 2009 through December 2019. The first is the universe of bid-level records from the BEC electronic procurement platform for SES/SP purchases of medical supplies (BEC Group 65). Each observation is a purchase-offer-item (POI), recording the reference price set during the planning phase, the winning bid price, the number of bidding firms, the quantity ordered, the modality (*pregão eletrônico* or *convite*), and the tender outcome.

The second is a purchase-type classifier that assigns each POI to one of three regimes: ordinary, administrative-request, or litigated. Brazilian law requires the tender notice for any urgent purchase to reference the underlying court order or administrative-channel request, and the classifier exploits this regulatory disclosure through a regular-expression algorithm applied to free-text notices. We validate it on a stratified hand-labeled sample of 500 notices (∼167 per predicted class); detailed metrics are in the Online Appendix.[4]

The third is the SES/SP S-CODES database of health-related litigation, which provides SUS-formulary status and injunction enforcement data for each litigated request.

The binary outcome *tender success* equals one if the POI closes with an accepted winning bid within its procurement order and zero otherwise; re-tendered items enter as new POIs and are not counted retroactively as successes. All continuous variables are winsorized at the 1st and 99th percentiles; robustness to alternative winsorization is in the Appendix.

3.2. Sample and descriptive patterns

The raw data comprise 479,330 POI observations across 33,144 distinct items, 51,059 purchase orders, and 100 PBUs (Online Appendix Table A.1). Our analysis sample restricts to items for which at least one ordinary and at least one litigated purchase is observed, ensuring within-item comparability; this yields 226,305 observations covering 3,856 items.[5] Price regressions further restrict to winning bids (196,995 observations).

Table 2 reports sample means by purchase type on winning bids. Three

[4]The regex relies on literal court-order references whose presence is independent of the price outcomes we estimate, so any residual misclassification attenuates rather than biases our coefficients.

[5]Sample sizes vary slightly across tables due to missing values in specific dependent variables and the restriction to winners for price regressions.

patterns anticipate the empirical results. *Higher litigated prices.* Log negotiated prices are higher in litigated than in ordinary purchases (1.050 vs. 1.596; item fixed effects absorb the level mix). *Larger administrative orders.* Administrative orders are markedly larger than litigated ones (794,534 vs. 61,903 units on average), the bulk-discount disadvantage that drives the mechanical channel in Section 5. *Thinner urgent competition with higher success.* The number of bidding firms is lower in urgent than in ordinary tenders (2.191 for litigated vs. 3.159 for ordinary), yet tender success rates are higher in urgent regimes (90.7% litigated vs. 84.1% ordinary)—officials accept worse terms to ensure compliance.

Table 2: Sample means by purchase type.

	Ordinary	Administrative	Litigated
Observations (winners)	343,393	16,985	46,212
Log reference price	2.109	0.960	1.578
Log negotiated price	1.596	0.543	1.050
Quantity (units)	56,280	794,534	61,903
Number of bidding firms	3.159	2.394	2.191
Tender success rate (%)	84.1	85.7	90.7

Notes: Means computed on winners (POIs with an accepted winning bid). Reference and negotiated prices in logs; quantity in units; tender success rate is the share of POIs that close with an accepted winning bid. Sample period: 2009–2019. Source: BEC-SP and SES/SP S-CODES.

3.3. The under-the-gun subsample

For the under-the-gun analysis, we construct a subsample of urgent purchases (administrative and litigated) for items with both regimes present, yielding 56,803 observations (Online Appendix Table A.2). The tightest specifications identify off item×year-month cells in which both regimes coexist: of 40,966 such cells, 4,402 (11%) carry within-cell variation. The remaining cells are absorbed as singletons under the fixed-effects design. Both-types cells are not a representative subsample—they tilt toward higher-volume, more-reordered medications (Online Appendix Table A.6); the bounded estimate therefore speaks to the more liquid items in the urgent panel, a restriction we revisit in the welfare bound. Online Appendix complements the balance with geographic evidence on the spatial distribution of administrative demand, which mirrors the litigation patterns of Section 2.

4. Empirical Strategy

We exploit two within-item contrasts. The urgent-vs-ordinary contrast identifies the cost margin associated with the urgent regime, conditional on item, time, and PBU fixed effects. The under-the-gun (UTG) contrast, restricted to the urgent panel, identifies the cost margin associated with the sanction regime, conditional on selection into the administrative channel that we bound formally.

4.1. Specifications

The urgent-vs-ordinary premium is

$$y_{i,g,t} = \beta \cdot \text{Urgent}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \tag{1}$$

where $y_{i,g,t}$ is log negotiated price, log reference price, log number of firms, or the success indicator for purchase order i of item g at time t; γ_g, λ_t, δ_b are item, time, and PBU fixed effects. We report four specifications with progressively richer fixed effects. The under-the-gun coefficient restricts to the urgent panel (admin or litigated):

$$\ln(\text{Price}_{i,g,t}) = \beta \cdot \text{Admin}_{i,g,t} + \gamma_g + \lambda_t + \delta_b + \varepsilon_{i,g,t}, \tag{2}$$

estimated as (i) a naive coefficient, (ii) a Manski-Lee bounded coefficient, and (iii) a within firm-buyer-item triple coefficient that isolates same-firm pricing from supplier composition. Standard errors cluster at the PBU level (99 clusters), with wild cluster bootstrap as a robustness check.

4.2. Identification: urgent vs. ordinary

Three institutional features support strict exogeneity of urgency conditional on item-time-PBU fixed effects. First, court orders and administrative requests originate from patient health needs and from external assessments by judges or the SES/SP scientific committee, not from the procurement process itself. Second, urgent demand correlates poorly with PBU mismanagement: only 4% of claims arise from stock shortages or service failures. Third, the urgency regime is determined outside the purchasing unit, not by it. We read the estimate as a within-item cost margin associated with the urgent regime, conditional on the institutional features above.

A Borusyak-Jaravel-Spiess (2024) imputation event-study around each item's first court order provides a dynamic check (Online Appendix). Pre-period coefficients are economically small (max absolute pre-period coefficient $= 0.041$); post-period coefficients rise from 0.052 at $t = 0$ to 0.147 at $t = +5$. A Rambachan-Roth (2023) honest sensitivity test gives a breakdown linear-extrapolation parameter at which the $t = 0$ confidence interval just contains zero of $M = 0.018$. The cross-sectional FE estimate is the primary object; the BJS event-study is supportive dynamic evidence rather than a stand-alone identification claim.

4.3. Identification: under-the-gun

Litigated and administrative urgent purchases face identical planning constraints (Section 2) and differ only in penalty exposure. Selection into the administrative channel is non-random: a scientific committee admits cases on cost-effectiveness criteria. The committee's first-stage probit on pre-period item characteristics rejects higher pre-period log reference price (-0.051, $p < 0.001$) and SUS-formulary status (-0.265, $p < 0.001$): admin systematically over-represents items that would have been priced lower under any regime, so the naive UTG comparison inflates the sanction effect.

We bound this wedge under monotone selection (Lee 2009) by trimming admin observations within item×year×PBU strata where the admin sample exceeds the litigated count. The trimming proportion $p_{\text{trim},s} = \max\{0, (n_{\text{adm},s} - n_{\text{lit},s})/n_{\text{adm},s}\}$ is the share of admin observations within stratum s that, under the Lee monotone-acceptance assumption, would not have appeared in the litigated counterfactual. Trimming admin from the upper tail of the within-stratum price distribution yields the lower bound on the admin-minus-lit log-price gap; trimming from the lower tail yields the upper bound. Average trimming intensity is 26.9% of admin observations where applicable; 12,038 of 44,654 strata trigger trimming. A parametric Heckman correction is non-identified in this design (Online Appendix A.5); the Lee bound is the primary selection-corrected object.

We do not estimate a structural counterfactual price under hypothetical sanction removal: that object is unidentified in our setting because the administrative subsample is selected on item economics. We instead provide a bounded decomposition of where the between-regime cost margin originates.

5. Results

5.1. Three motivating patterns

Within the same item, urgent purchases carry negotiated prices 5.4% higher than ordinary purchases and attract -5.4% fewer bidding firms; reference prices set during the planning phase rise by 2.7% ($p < 0.10$), attenuating sharply once buyer fixed effects absorb buyer-level heterogeneity. Despite worse conditions, urgent tenders *succeed* more often: 2.1 pp ($p < 0.01$). Officials accept worse terms to ensure compliance rather than cancel; the urgency premium is not selection on easier purchases.

Inside urgent procurement, the naive Admin coefficient is 29.5% ($p < 0.05$; wild cluster bootstrap $p_{\text{boot}} = 0.0080$). The SES/SP scientific committee admits cases on cost-effectiveness criteria; under Manski-Lee monotone restrictions on its selection rule (Section 4), the selection-corrected gap falls in $[15.9\%, 21.1\%]$ (Online Appendix A.4). The remainder of this section identifies through which mechanism that bounded gap operates.[6]

Table 3: Negotiated Prices

	(1)	(2)	(3)	(4)
Panel A: Total Effect				
Urgent Purchase	0.159***	0.151***	0.053***	0.056***
	(0.042)	(0.038)	(0.016)	(0.016)
Panel B: Direct Effect				
Urgent Purchase	0.075	0.060	0.030*	0.030*
	(0.045)	(0.042)	(0.017)	(0.017)
Log Quantity	-0.321***	-0.325***	-0.392***	-0.393***
	(0.026)	(0.026)	(0.029)	(0.029)
Item FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	No
Year-Month FE	No	No	No	Yes
PBU FE	No	No	Yes	Yes
Observations	196,886	196,886	196,883	196,883
Within R^2 (A)	0.003	0.003	0.000	0.000
Within R^2 (B)	0.315	0.327	0.358	0.359

Notes: Dependent variable: log negotiated price. Sample: winners only. Panel A reports the total effect; Panel B controls for log quantity. Winsorized at 1%/99%. Standard errors clustered at the PBU level in parentheses. *** p<0.01, ** p<0.05, * p<0.1.

[6] The 5.4% within-item premium sits at the lower end of within-buyer procurement-cost dispersion documented in comparable settings: Bandiera-Prat-Valletti 2009 10–20%; Best, Hjort & Szakonyi 2023 20–40%; Bosio et al. 2022 15–25%; Coviello-Guglielmo-Spagnolo 2018 5–12%. Tighter within-buyer within-item identification is consistent with a smaller headline number.

5.2. Same firm, same buyer, same item: no within-firm markup

Restricting to firm-buyer-item triples observed under both the litigated and the administrative regime (1,206 triples covering 4,573 observations), the Admin coefficient is 0.035 (SE 0.041): statistically indistinguishable from zero (95% bootstrap CI [-6.6%, 11.0%]). The same firm prices the same item to the same buyer at the same level under either regime; suppliers who already serve the buyer's urgent procurement do not extract a markup when the official is sanction-exposed.

This is the empirical pivot of the paper. The standard accountability-distortion intuition (Prendergast, 2007) predicts that sanction-exposed officials face higher per-unit prices because suppliers extract a delivery-risk premium; in the bulk of our data, that mechanism is absent.

Robustness across subsamples.. Re-fitting the within firm-buyer-item specification on five subsamples (Online Appendix Table A.7), the null is robust on the deepest markets—above-median quantity (-0.005, SE 0.044), SUS-formulary items (-0.001, SE 0.026), and the later period 2014–2019 (-0.038, SE 0.052). The null breaks on thin-supplier slices: below-median quantity items ($+0.066$, $p < 0.05$) and the earlier period 2009–2013 ($+0.117$, $p < 0.01$), where the incumbent does extract a markup exactly as Prendergast (2007) predicts when the supplier has outside-option leverage. The headline within-firm null is a deep-market result; the thin-market deviation locates where the delivery-risk-premium prediction does bite.

Table 4: UTG reconciliation: observed log-price gap decomposed into mechanical bulk-discount prediction, within firm-buyer-item offset, and supplier composition residual. 1,206 triples observed in both regimes.

Component	Log-points (admin minus lit)
Observed UTG (admin minus lit, log price)	-0.259
Mechanical C1: bulk-discount x qty gap	-0.397
Within firm-buyer-item offset	0.035
Supplier composition residual	0.104

Identity: observed UTG = mechanical C1 + within-firm offset + composition residual. Within-fi

5.3. The gap is sourcing, not pricing: decomposition and heterogeneity

We decompose the observed log-price gap (admin minus litigated, -0.259) into three components linked by an arithmetic identity:

$$\underbrace{\Delta y_{\text{obs}}}_{-0.259} = \underbrace{\beta_{\text{bulk}} \cdot \Delta \ln Q_{\text{adm-lit}}}_{\text{mechanical: } -0.397} + \underbrace{\beta_{\text{within-firm}}}_{\text{within-firm: } +0.035} + \underbrace{\beta_{\text{composition}}}_{\text{residual: } +0.103}.$$

Three readings follow, with 95% cluster-bootstrap CIs (PBU, 499 replicates). *Bulk-discount channel identified and large*: -32.8% ([-51.2%, -5.2%]). Order-size differences alone would make admin cheaper by 32.8%, more than the 23% actually observed. *Within-firm channel identified and zero*: 3.5% ([-6.6%, 11.0%]). *Residual channel imprecisely identified*: 10.9% on the point estimate, ([-16.0%, 39.6%]) on the bootstrap, and inherits variance from the other three bars by construction. The residual loads what the identity attributes neither to scale nor to within-firm pricing: equilibrium changes in *which* firm wins, match-quality shifts, and any other margin not separately identified in this design.

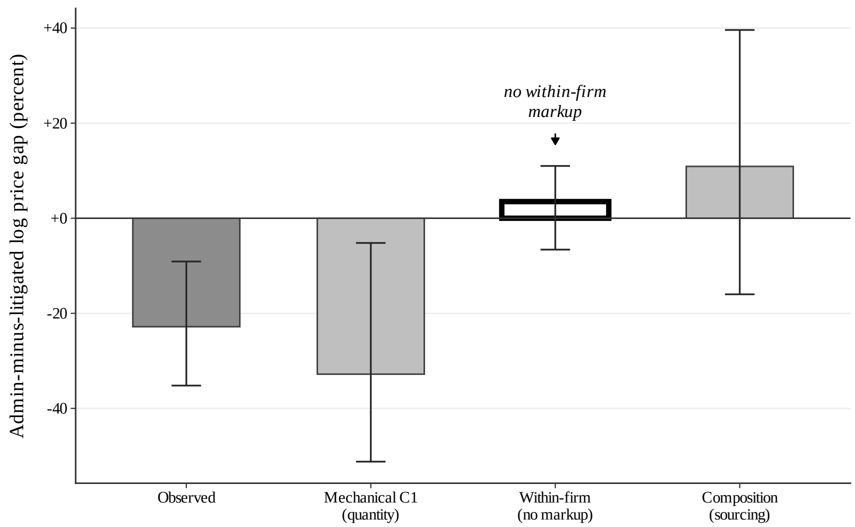

Figure 1: Sourcing vs. pricing: the within-firm null is the empirical pivot.
Notes: Bars give the four components of the admin-minus-lit log-price identity in Table 4; whiskers are 95% cluster bootstrap on PBU with 499 replicates. The within firm-buyer-item bar (white, heavy outline) brackets zero: same firm, same buyer, same item, same price across regimes. The mechanical bulk-discount channel is identified and over-explains the observed gap; the residual composition channel loads the remainder but is imprecisely identified.

Direct mechanism: winners shift, prices don't.. The composition residual in the decomposition above is identified by elimination; we now provide direct evidence that the sourcing margin loads it. Among the 2,134 item-buyer pairs observed under both regimes, the winner-firm sets overlap only modestly: mean Jaccard similarity is 0.268, 48.5% of pairs have *no* winner in common across regimes, and the modal winner differs across regimes in 70.2% of pairs. Conditional on the same item and buyer, switching from administrative to litigated procurement reorganizes the supplier set far more than it reorganizes incumbent pricing. Table 5 reports the full distribution.

Table 5: Winner switching across regimes for the same item-buyer pair.

Item-buyer pairs (both regimes)	2,134
Mean distinct winners, admin	2.05
Mean distinct winners, litigated	1.99
Mean Jaccard similarity, winner sets	0.268
Pairs with any winner overlap (%)	51.5
Pairs with NO winner overlap (%)	48.5
Pairs with same modal winner (%)	29.8

Notes: Sample of buyer×item pairs with at least one administrative and one litigated urgent purchase. Jaccard similarity is the ratio of the cardinality of the intersection to the cardinality of the union of the two regimes' winning-firm sets. "Modal winner" is the most frequent winning firm within each regime for a given pair. Source: BEC-SP, 2009–2019.

Where the cost bites confirms the sourcing reading.. Splitting items by mean number of participating firms (median 2.071), competitive items show a 14.4% urgency premium with 8.5% surviving firm fixed effects (39% attenuation). Concentrated items show no detectable premium (-1.0%, not significant). Urgency erodes competitive pressure where there is competitive pressure to erode—the heterogeneity matches the sourcing channel, not a pricing channel.

5.4. Falsification: items never litigated

The urgency premium exists only inside the litigation pool. Running the preferred specification on items that received zero litigated purchases across 2009–2019 (restricted to those with both ordinary and administrative purchases) returns placebo coefficients of -0.020 (SE 0.032) for negotiated prices and -0.031 (SE 0.045) for reference prices. Both are economically small and statistically insignificant (Online Appendix Table A.9).

6. Conclusion

Courts that compel governments to buy medications save individual lives. They also raise public procurement costs by 5.4% per purchase across the full sample and attract -5.4% fewer bidders. Once selection into the SES/SP administrative channel is bounded, the under-the-gun gap is $[15.9\%, 21.1\%]$.

The empirical signature of this inefficiency is the *absence* of a within-firm markup, not its presence. Within firm-buyer-item triples observed under both regimes, the same firm prices the same item to the same buyer at a statistically indistinguishable level (3.5%, bootstrap CI [-6.6%, 11.0%]). Suppliers who already serve the buyer do not extract a delivery-risk premium when the official faces sanctions. The cost margin is sourcing under fragmentation, not pricing under sanctions.

Welfare bound.. Applied to \$300 M of annual litigated spending in São Paulo and an admissibility calibration of 50.0%, the bounded UTG implies a per-unit-price welfare cost of \$27.8 M per year (Lee range $[\$23.9\ M, \$31.7\ M]$). This is a per-unit-price margin and excludes welfare losses from constrained order sizes (the bulk-discount channel itself), reduced bidder participation, and officials' search time diverted from ordinary procurement.

Policy implications.. Two reforms address the two channels we identify. *Demand side*: expanding the administrative-request mechanism eliminates within-urgent sanction exposure for the admissible share. Because the within-firm pricing component is zero, the gain is realized through demand aggregation; the binding parameter is committee capacity, not legal reform. *Supply side*: framework agreements for repeat-purchase commodity-like medications address fragmentation directly by allowing aggregation across urgent and ordinary requests. We do not attach a recovery point estimate to the supply-side channel: any number requires a calibration anchor on item eligibility we are not positioned to provide.

Boundary and outlook.. Results are estimated within São Paulo's BEC platform, on pharmaceutical procurement in a single state, in 2009–2019. We do not extrapolate to other states, to non-pharmaceutical procurement, or to post-reform institutional arrangements. The bounded estimate applies within the urgent panel and tilts toward higher-volume formulary medications (Online Appendix Table A.6); the welfare figure assumes that admin admissibility can be expanded without affecting the per-unit-price gap, an assumption the manuscript does not test. The broader implication is methodological: standard accountability designs are rationalized on the prediction that sanctions discipline prices through suppliers' delivery-risk premia. Our null on that channel reveals the binding margin under one-sided accountability is the sourcing pattern itself; the policy lever shifts accordingly from contract design to demand aggregation.

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Online Appendix

This appendix collects the tables and figures referenced from the main text. Section A.1 reports sample descriptives and balance. Section A.2 collects the selection-bound and robustness exhibits. Section A.3 reports the BJS event study with Rambachan-Roth honest sensitivity. Section A.4 reports the placebo on never-litigated items. Section A.5 documents the regex classifier validation.

A.1 Sample and Balance

Table A.1: Descriptive Statistics by Purchase Type

	Ordinary		Administrative		Litigated		Diff	Diff
	Mean	(SD)	Mean	(SD)	Mean	(SD)	(O–L)	(A–L)
Panel A: Levels								
Reference Price	909.17	(5,959.41)	226.38	(2,257.35)	370.39	(2,540.20)	538.78***	-144.01***
							[0.000]	[0.000]
Negotiated Price	477.51	(2,915.88)	148.03	(1,177.52)	260.25	(1,561.81)	217.26***	-112.22***
							[0.000]	[0.000]
Quantity	31,487.15	(163,507.29)	117,146.00	(356,962.37)	9,646.19	(82,683.10)	21,840.96***	107,499.81***
							[0.000]	[0.000]
N. Bidding Firms	3.17	(1.93)	2.43	(1.35)	2.20	(1.19)	0.97***	0.23***
							[0.000]	[0.000]
Panel B: Log Transformations								
Log Reference Price	0.926	(2.723)	0.909	(2.541)	1.557	(2.446)	-0.632***	-0.648***
							[0.000]	[0.000]
Log Negotiated Price	0.319	(2.862)	0.480	(2.604)	1.024	(2.568)	-0.705***	-0.544***
							[0.000]	[0.000]
Log Quantity	6.947	(2.678)	7.472	(3.160)	6.087	(2.083)	0.860***	1.385***
							[0.000]	[0.000]
Log N. Firms	0.984	(0.590)	0.747	(0.529)	0.665	(0.493)	0.319***	0.081***
							[0.000]	[0.000]
Panel C: Tender Characteristics								
Successful Tender (%)	0.859	(0.348)	0.869	(0.338)	0.908	(0.288)	-0.050***	-0.040***
							[0.000]	[0.000]
Observations	136,250		15,371		45,351			

Notes: Sample restricted to items with at least one ordinary and one litigated purchase. Winners only for price and quantity variables. Variables winsorized at 1%/99%. Stars on differences from Welch t-tests; p-values in brackets. *** p<0.01, ** p<0.05, * p<0.1.

Table A.2: Balance Table: Administrative vs Litigated (Urgent Purchases)

	Administrative		Litigated		Diff	p-value
	Mean	(SD)	Mean	(SD)	(A–L)	
Panel A: Procurement Outcomes						
Reference Price	226.38	(2,257.35)	232.56	(1,432.18)	-6.18	[0.752]
Negotiated Price	148.03	(1,177.52)	174.79	(1,052.56)	-26.75**	[0.013]
Quantity	117,146.00	(356,962.37)	9,306.87	(80,252.25)	107,839.13***	[0.000]
Log Reference Price	0.91	(2.54)	1.43	(2.32)	-0.52***	[0.000]
Log Negotiated Price	0.48	(2.60)	0.89	(2.44)	-0.41***	[0.000]
Log Quantity	7.47	(3.16)	6.18	(1.99)	1.29***	[0.000]
Panel B: Market Structure						
N. Bidding Firms	2.427	(1.351)	2.155	(1.120)	0.272***	[0.000]
Log N. Firms	0.747	(0.529)	0.650	(0.483)	0.097***	[0.000]
Successful Tender (%)	0.869	(0.338)	0.914	(0.280)	-0.045***	[0.000]
Panel C: Purchase Characteristics						
Electronic Auction	0.985	(0.123)	0.941	(0.236)	0.044***	[0.000]
Observations	15,371		41,491			

Notes: Sample restricted to urgent purchases (administrative and litigated) for items with both types present. Winners only for price/quantity. Variables winsorized at 1%/99%. p-values from Welch t-tests in brackets. *** p<0.01, ** p<0.05, * p<0.1.

Table A.3: Within-Item Balance: Administrative vs. Litigated

Covariate	Mean (Admin)	Mean (Lit)	Raw diff. (A–L)	Within-item diff. Coef	Within-item diff. (SE)
SUS basic (MEDICAMENTO flag)	0.718	0.892	-0.174	0.000^{NA}	(NaN)
Electronic auction (pregão)	0.984	0.941	0.043	0.041^{**}	(0.017)
Log quantity (bulk size signal)	7.642	6.180	1.462	0.866^{*}	(0.502)
Log reference price	0.835	1.401	-0.566	-0.339^{***}	(0.093)
Successful tender	0.869	0.914	-0.045	-0.024	(0.022)
Late period (year $\geq$ 2014)	0.565	0.623	-0.058	-0.045	(0.041)
Large PBU (above-median)	0.895	0.810	0.085	0.083	(0.073)
Observations			63,426		
Items			2,370		

Notes: Within-item balance between administrative ($Admin = 1$) and litigated ($Admin = 0$) urgent purchases, restricted to items with both types present. Raw difference is the unconditional mean gap. Within-item coefficient is β from $x_{i,g,t} = \beta \, \text{Admin}_{i,g,t} + \gamma_g + \varepsilon_{i,g,t}$, where γ_g is an item fixed effect and standard errors are clustered at the PBU level. A coefficient close to zero means that, within a given item, the administrative and litigated draws are balanced on the covariate. Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.2 Selection Bounds and Robustness

Manski-Lee bounds.. Table A.4 reports the bounded UTG coefficient under monotone selection. Trimming admin observations within item×year×PBU strata where the admin sample exceeds the litigated count yields a bounded gap in $[15.9\%, 21.1\%]$ against a naive coefficient of 29.5%. Average trimming intensity is 26.9% of admin observations across 12,038 of 44,654 strata.

Heckman selection model (non-identified).. Table A.5 reports a parametric Heckman correction with the same first-stage probit (pre-period log reference price, mean quantity, mean number of bidders, and order count as the exclusion restriction; SUS-formulary indicator). The probit on $N = 5{,}351$ observations recovers negative loadings on pre-period reference price and SUS status. The inverse Mills ratio is collinear with the item, year, and PBU fixed effects in the second stage; the resulting Admin coefficient (0.817, SE 4.278) is uninformative. The Lee bound is the primary selection-corrected object.

Both-types-cell representativeness.. Table A.6 compares the 5,581 item×year-month cells that contain both administrative and litigated urgent purchases

Table A.4: Manski-Lee bounds on the under-the-gun coefficient.

	log negotiated price		implied
Specification	Coef. on Admin	SE	Lit-vs-Admin (%)
Naive UTG (item + year + PBU FE)	-0.259	0.092	29.530
Lee lower bound (admin top-trim)	-0.192	0.095	15.924
Lee upper bound (admin bottom-trim)	-0.148	0.097	21.123

Trimming applied within item*times*year*times*PBU strata where the admin sample exceeds the li

Table A.5: Heckman selection-corrected UTG estimate.

Term	Estimate
First stage: pre-period log ref price	-0.051
First stage: pre-period mean qty	0.000
First stage: pre-period mean firms	0.065
First stage: pre-period n orders (excl)	0.000
First stage: SUS formulary	-0.265
Second stage: Admin coef	0.817
Second stage: inverse Mills ratio	-0.653
Second stage: implied lit-over-admin (%)	-55.841
Heckit ML: ρ	1.000

First stage: probit of admin on pre-period item characteristics. pre_n_orders is excluded from t

(the within-cell variation that identifies the tightest specification) with the 11,972 singleton cells. Both-types cells systematically have lower mean log reference and negotiated prices, larger quantities, slightly fewer bidders, and a higher SUS-formulary share. The bounded UTG estimate therefore speaks to the more liquid, formulary-anchored items in the urgent panel.

Within-firm null robustness.. Table A.7 re-fits the within firm-buyer-item Admin specification on five subsamples constructed from the triple sample (1,206 triples). The null is robust on above-median quantity, SUS-formulary, and the later period; it breaks on below-median quantity (+6.6%, $p < 0.05$) and the earlier period (+11.7%, $p < 0.01$). The pattern locates the deep-market versus thin-market boundary between the BPV sourcing reading and the Prendergast within-firm-pricing reading.

Table A.6: Representativeness of both-types cells: cells with both administrative and litigated urgent purchases (5,581 cells, 31.8%) vs. singleton cells (11,972).

Variable	Both-types cells	Singleton cells
Log reference price	1.233	1.761
Log negotiated price	0.703	1.326
Log quantity	6.560	6.324
N. bidding firms	2.181	2.404
SUS formulary share	0.867	0.725
Reverse-auction (pregao) share	0.955	0.952

Means computed at the observation level within each cell type. p-values from two-sample t-tests.

Wild cluster bootstrap.. Table A.8 reports Rademacher wild cluster bootstrap inference on the UTG coefficient (999 replicates, PBU clustering, restricted-null residuals). The preferred-FE Admin coefficient is significant under both asymptotic and bootstrap inference ($p_{\text{boot}} = 0.0080$, 95% CI in log-points $[-0.469, -0.039]$); the item×year-month FE specification is borderline ($p_{\text{boot}} = 0.0390$), as expected given the thin within-cell sample.

Table A.7: Within firm-buyer-item null: robustness across subsamples.

Subsample	$\hat{\beta}_{\text{Admin}}$	SE	N
All triples (baseline)	0.035	0.041	4,573
Above-median quantity	-0.005	0.044	2,188
Below-median quantity	0.066***	0.025	2,027
SUS-formulary items	-0.001	0.026	2,540
Non-formulary items	0.101	0.064	2,033
Earlier period	0.117***	0.037	2,553
Later period	-0.038	0.052	1,492

Notes: Each row reports the within firm-buyer-item triple Admin coefficient on log negotiated price, with FBI and year fixed effects, PBU-clustered SEs. Subsamples are constructed from the triple sample (1,206 triples). Significance: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A.8: Wild cluster bootstrap inference on the UTG coefficient.

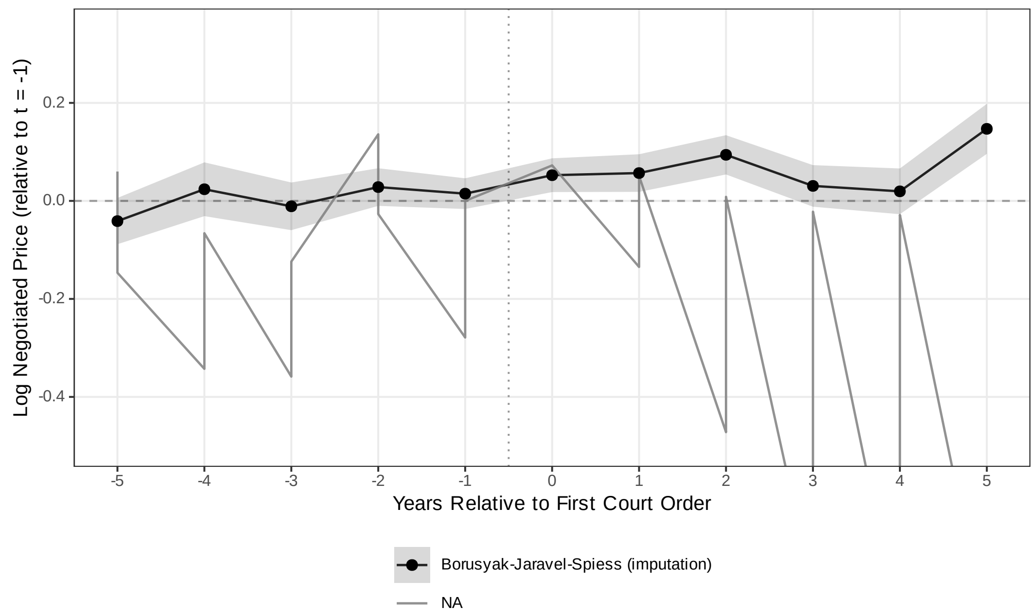
Specification	$\hat{\beta}$	Asy. SE	t	p_{boot}	95% CI (log-points)
Preferred (item + year + PBU)	-0.259	0.092	-2.80	0.008	$[-0.469, -0.039]$
Tightest (item×year-month + PBU)	-0.283	0.116	-2.44	0.039	$[-0.549, -0.001]$

Notes: Rademacher wild cluster bootstrap on the Admin coefficient with $B = 999$ replicates and restricted-null residuals; clustering at the PBU level. Negative coefficients mean litigated purchases are more expensive than administrative ones, in log-points. Source: `analysis/44_wild_bootstrap.R`.

A.3 Event Study with Honest Sensitivity

We re-estimate the dynamic response of log negotiated price (and log quantity) to the first court order for each item using the imputation estimator of Borusyak et al. (2024) (BJS) and the ATT(g, t) estimator of Callaway and Sant'Anna (2021) (CS). The panel is item-year, winner-only; treatment is the calendar year of the first litigated purchase for each item; the never-litigated set is the CS control group. BJS pre-period coefficients are economically small (< 0.041 in absolute value); post-period coefficients rise from $+0.052$

at $t = 0$ to $+0.147$ at $t = +5$. Within the same item, order quantity drops sharply at the year of the first court order (-27.8% at $t = 0$) and recovers slowly (-0.3% at $t = +5$), tracing the fragmentation channel in real time.



referred (shaded 95% CI). CS with never-treated control is noisier because never-litigated items are a systematically different comparison group.

Figure A.1: Log negotiated price around first court order: honest-DiD estimators. *Notes:* BJS shown with shaded 95% CI; CS with never-treated control; TWFE as a pre-reform benchmark. CS is noisier because never-litigated items are systematically different (routine, well-stocked).

Rambachan-Roth honest sensitivity.. Figure A.2 reports the BJS event-study coefficients with a Rambachan-Roth honest sensitivity overlay (Rambachan and Roth, 2023). Pre-period maximum absolute coefficient is 0.041; the $t = 0$ effect is 0.052 (SE 0.018); the breakdown linear-extrapolation parameter at which the $t = 0$ confidence interval just contains zero is $M = 0.018$. The $t = 0$ effect is significant under parallel trends but does not survive the linear extrapolation of the observed pre-period maximum—we honor this in the main text by treating the cross-sectional fixed-effects estimate as primary and the BJS event study as supportive, not as a stand-alone identification claim.

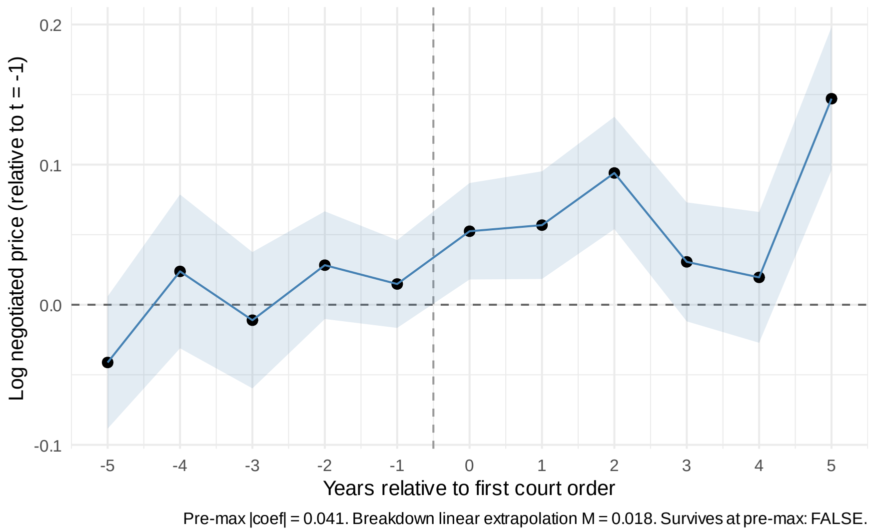


Figure A.2: BJS event study around first court order with Rambachan-Roth sensitivity diagnostic.

A.4 Falsification: Items Never Litigated

Running the preferred specification on items with zero litigated purchases across 2009–2019 (restricted to those with both ordinary and administrative purchases) yields placebo coefficients of -0.020 (SE 0.032) for negotiated prices and -0.031 (SE 0.045) for reference prices. Both are economically small and statistically insignificant.

Table A.9: Placebo Test: Items Never Subject to Litigation

	bid_price_log		bid_price_ref_log	
	(1)	(2)	(3)	(4)
Urgent Purchase	-0.0204	0.0513***	-0.0306	0.0305**
	(0.0316)	(0.0153)	(0.0447)	(0.0141)
Observations	39,283	196,883	46,874	226,278
R^2	0.83364	0.86796	0.81121	0.85183
Within R^2	7.03×10^{-6}	0.00022	1.73×10^{-5}	7.43×10^{-5}
item_id fixed effects	✓	✓	✓	✓
year_n fixed effects	✓	✓	✓	✓
pbu_id fixed effects	✓	✓	✓	✓

Placebo sample: items with zero litigated purchases across 2009-2019. Main sample: items with at least one litigated and one ordinary purchase. DV: log price. Item + Year + PBU FE. SE clustered at PBU level.

A.5 Regex Classifier Validation

The purchase-type indicator is constructed by a regular-expression algorithm applied to public tender notices, which by Brazilian procurement law must reference any underlying court order. We draw a stratified random sample of 500 tender-notice subjects (∼167 per predicted class) and hand-label each as ordinary, administrative, or litigated. Hand-labeling of the stratified sample is in progress; F1 diagnostics will be reported in the replication archive accompanying this paper. The regex relies on textual markers present or absent independently of the price outcomes we estimate, so misclassification attenuates rather than biases our estimates.[7]

[7] The stratified sample is reproduced from `output/validation/validation_sample.csv`, generated by `analysis/33_regex_validation_sample.R`; F1 statistics are emitted by `34_regex_validation_f1.R` when the labeled CSV is finalized.